* 0000800000001 *

Cambridge Assessment International Education

Cambridge International AS & A Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

8986046942

MATHEMATICS **9709/11**

Paper 1 Pure Mathematics 1 **October/November 2024**

1 hour 50 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **20** pages. Any blank pages are indicated.

© UCLES 2024

[Turn over

1 In the expansion of $\left(kx+\frac{2}{x}\right)^4$, where k is a positive constant, the term independent of x is equal to 150.

Find the value of k and hence determine the coefficient of x^2 in the expansion. [4]

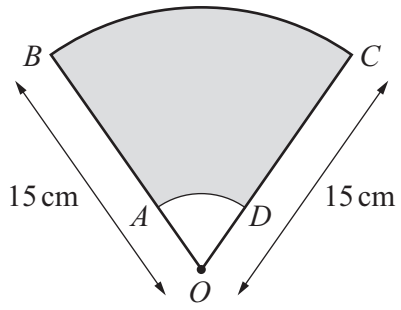
DO NOT WRITE IN THIS MARGIN

2 The curve $y = x^2 - \frac{a}{x}$ has a stationary point at $(-3, b)$.

Find the values of the constants a and b. [4]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

3



The diagram shows a sector of a circle, centre O, where $OB = OC = 15\,\text{cm}$. The size of angle BOC is $\frac{2}{5}\pi$ radians. Points A and D on the lines OB and OC respectively are joined by an arc AD of a circle with centre O. The shaded region is bounded by the arcs AD and BC and by the straight lines AB and DC. It is given that the area of the shaded region is $\frac{209}{5}\pi\,\text{cm}^2$.

Find the perimeter of the shaded region. Give your answer in terms of π. [5]

DO NOT WRITE IN THIS MARGIN

4 Show that the curve with equation $x^2 - 3xy - 40 = 0$ and the line with equation $3x + y + k = 0$ meet for all values of the constant k. [5]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

5 The equation of a curve is such that $\frac{dy}{dx} = 4x - 3\sqrt{x} + 1$.

(a) Find the x-coordinate of the point on the curve at which the gradient is $\frac{11}{2}$. [3]

(b) Given that the curve passes through the point (4, 11), find the equation of the curve. [4]

DO NOT WRITE IN THIS MARGIN

6 Circles C_1 and C_2 have equations

$$x^2 + y^2 + 6x - 10y + 18 = 0 \text{ and } (x-9)^2 + (y+4)^2 - 64 = 0$$

respectively.

(a) Find the distance between the centres of the circles. [4]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

P and Q are points on C_1 and C_2 respectively. The distance between P and Q is denoted by d.

(b) Find the greatest and least possible values of d. [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

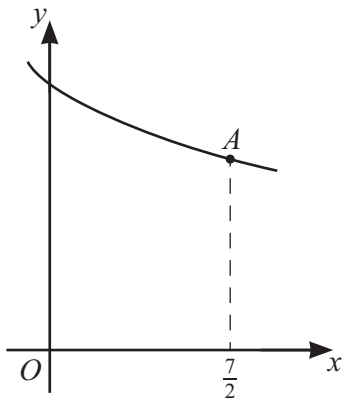
.....

.....

.....

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

7



The diagram shows part of the curve with equation $y = \frac{12}{\sqrt[3]{2x+1}}$. The point A on the curve has coordinates $\left(\frac{7}{2}, 6\right)$.

(a) Find the equation of the tangent to the curve at A. Give your answer in the form $y = mx + c$. [4]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

DO NOT WRITE IN THIS MARGIN

(b) Find the area of the region bounded by the curve and the lines $x = 0$, $x = \frac{7}{2}$ and $y = 0$. [4]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

8 **(a)** It is given that β is an angle between 90° and 180° such that $\sin\beta = a$.

Express $\tan^2\beta - 3\sin\beta\cos\beta$ in terms of a. [3]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(b) Solve the equation $\sin^2\theta + 2\cos^2\theta = 4\sin\theta + 3$ for $0° < \theta < 360°$. [5]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

9 The equation of a curve is $y = 4 + 5x + 6x^2 - 3x^3$.

(a) Find the set of values of x for which y decreases as x increases. [4]

DO NOT WRITE IN THIS MARGIN

(b) It is given that $y = 9x + k$ is a tangent to the curve.

Find the value of the constant k. [4]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

10 An arithmetic progression has first term 5 and common difference d, where $d > 0$. The second, fifth and eleventh terms of the arithmetic progression, in that order, are the first three terms of a geometric progression.

(a) Find the value of d. [3]

DO NOT WRITE IN THIS MARGIN

(b) The sum of the first 77 terms of the arithmetic progression is denoted by S_{77}. The sum of the first 10 terms of the geometric progression is denoted by G_{10}.

Find the value of $S_{77} - G_{10}$. [5]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

11 The function f is defined by $\mathrm{f}(x) = 3 + 6x - 2x^2$ for $x \in \mathbb{R}$.

(a) Express $\mathrm{f}(x)$ in the form $a - b(x - c)^2$, where a, b and c are constants, and state the range of f. [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(b) The graph of $y = \mathrm{f}(x)$ is transformed to the graph of $y = \mathrm{h}(x)$ by a reflection in one of the axes followed by a translation. It is given that the graph of $y = \mathrm{h}(x)$ has a minimum point at the origin.

Give details of the reflection and translation involved. [2]

.....

.....

.....

.....

.....

.....

.....

.....

.....

DO NOT WRITE IN THIS MARGIN

The function g is defined by $g(x) = 3 + 6x - 2x^2$ for $x \leqslant 0$.

(c) Sketch the graph of $y = g(x)$ and explain why g is a one-one function. You are **not** required to find the coordinates of any intersections with the axes. [2]

.....

.....

.....

(d) Sketch the graph of $y = g^{-1}(x)$ on your diagram in **(c)**, and find an expression for $g^{-1}(x)$. You should label the two graphs in your diagram appropriately and show any relevant mirror line. [4]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

Additional page

If you use the following lined page to complete the answer(s) to any question(s), the question number(s) must be clearly shown.

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

BLANK PAGE

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

BLANK PAGE

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

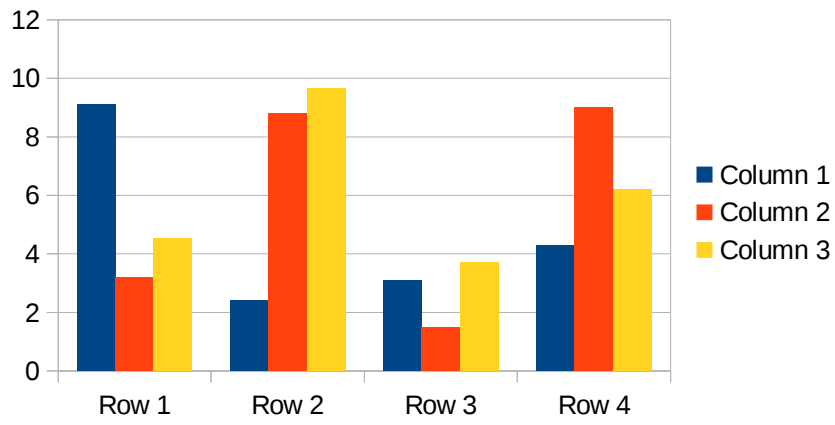
Lorem ipsum

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc ac faucibus odio.

Vestibulum neque massa, scelerisque sit amet ligula eu, congue molestie mi. Praesent ut varius sem. Nullam at porttitor arcu, nec lacinia nisi. Ut ac dolor vitae odio interdum condimentum. **Vivamus dapibus sodales ex, vitae malesuada ipsum cursus convallis. Maecenas sed egestas nulla, ac condimentum orci.** Mauris diam felis, vulputate ac suscipit et, iaculis non est. Curabitur semper arcu ac ligula semper, nec luctus nisl blandit. Integer lacinia ante ac libero lobortis imperdiet. *Nullam mollis convallis ipsum, ac accumsan nunc vehicula vitae.* Nulla eget justo in felis tristique fringilla. Morbi sit amet tortor quis risus auctor condimentum. Morbi in ullamcorper elit. Nulla iaculis tellus sit amet mauris tempus fringilla.

Maecenas mauris lectus, lobortis et purus mattis, blandit dictum tellus.

- **Maecenas non lorem quis tellus placerat varius.**
- *Nulla facilisi.*
- Aenean congue fringilla justo ut aliquam.
- Mauris id ex erat. Nunc vulputate neque vitae justo facilisis, non condimentum ante sagittis.
- Morbi viverra semper lorem nec molestie.
- Maecenas tincidunt est efficitur ligula euismod, sit amet ornare est vulputate.



In non mauris justo. Duis vehicula mi vel mi pretium, a viverra erat efficitur. Cras aliquam est ac eros varius, id iaculis dui auctor. Duis pretium neque ligula, et pulvinar mi placerat et. Nulla nec nunc sit amet nunc posuere vestibulum. Ut id neque eget tortor mattis tristique. Donec ante est, blandit sit amet tristique vel, lacinia pulvinar arcu. Pellentesque scelerisque fermentum erat, id posuere justo pulvinar ut. Cras id eros sed enim aliquam lobortis. Sed lobortis nisl ut eros efficitur tincidunt. Cras justo mi, porttitor quis mattis vel, ultricies ut purus. Ut facilisis et lacus eu cursus.

In eleifend velit vitae libero sollicitudin euismod. Fusce vitae vestibulum velit. Pellentesque vulputate lectus quis pellentesque commodo. Aliquam erat volutpat. Vestibulum in egestas velit. Pellentesque fermentum nisl vitae fringilla venenatis. Etiam id mauris vitae orci maximus ultricies.

Cras fringilla ipsum magna, in fringilla dui commodo a.

	Lorem ipsum	Lorem ipsum	Lorem ipsum
1	In eleifend velit vitae libero sollicitudin euismod.	Lorem	
2	Cras fringilla ipsum magna, in fringilla dui commodo a.	Ipsum	
3	Aliquam erat volutpat.	Lorem	
4	Fusce vitae vestibulum velit.	Lorem	
5	Etiam vehicula luctus fermentum.	Ipsum	

Etiam vehicula luctus fermentum. In vel metus congue, pulvinar lectus vel, fermentum dui. Maecenas ante orci, egestas ut aliquet sit amet, sagittis a magna. Aliquam ante quam, pellentesque ut dignissim quis, laoreet eget est. Aliquam erat volutpat. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Ut ullamcorper justo sapien, in cursus libero viverra eget. Vivamus auctor imperdiet urna, at pulvinar leo posuere laoreet. Suspendisse neque nisl, fringilla at iaculis scelerisque, ornare vel dolor. Ut et pulvinar nunc. Pellentesque fringilla mollis efficitur. Nullam venenatis commodo imperdiet. Morbi velit neque, semper quis lorem quis, efficitur dignissim ipsum. Ut ac lorem sed turpis imperdiet eleifend sit amet id sapien.

Lorem ipsum dolor sit amet, consectetur adipiscing elit.

Nunc ac faucibus odio. Vestibulum neque massa, scelerisque sit amet ligula eu, congue molestie mi. Praesent ut varius sem. Nullam at porttitor arcu, nec lacinia nisi. Ut ac dolor vitae odio interdum condimentum. Vivamus dapibus sodales ex, vitae malesuada ipsum cursus convallis. Maecenas sed egestas nulla, ac condimentum orci. Mauris diam felis, vulputate ac suscipit et, iaculis non est. Curabitur semper arcu ac ligula semper, nec luctus nisl blandit. Integer lacinia ante ac libero lobortis imperdiet. Nullam mollis convallis ipsum, ac accumsan nunc vehicula vitae. Nulla eget justo in felis tristique fringilla. Morbi sit amet tortor quis risus auctor condimentum. Morbi in ullamcorper elit. Nulla iaculis tellus sit amet mauris tempus fringilla.

Maecenas mauris lectus, lobortis et purus mattis, blandit dictum tellus.

Maecenas non lorem quis tellus placerat varius. Nulla facilisi. Aenean congue fringilla justo ut aliquam. Mauris id ex erat. Nunc vulputate neque vitae justo facilisis, non condimentum ante sagittis. Morbi viverra semper lorem nec molestie. Maecenas tincidunt est efficitur ligula euismod, sit amet ornare est vulputate.

In non mauris justo. Duis vehicula mi vel mi pretium, a viverra erat efficitur. Cras aliquam est ac eros varius, id iaculis dui auctor. Duis pretium neque ligula, et pulvinar mi placerat et. Nulla nec nunc sit amet nunc posuere vestibulum. Ut id neque eget tortor mattis tristique. Donec ante est, blandit sit amet tristique vel, lacinia pulvinar arcu. Pellentesque scelerisque fermentum erat, id posuere justo pulvinar ut. Cras id eros sed enim aliquam lobortis. Sed lobortis nisl ut eros efficitur tincidunt. Cras justo mi, porttitor quis mattis vel, ultricies ut purus. Ut facilisis et lacus eu cursus.

In eleifend velit vitae libero sollicitudin euismod.

Fusce vitae vestibulum velit. Pellentesque vulputate lectus quis pellentesque commodo. Aliquam erat volutpat. Vestibulum in egestas velit. Pellentesque fermentum nisl vitae fringilla venenatis. Etiam id mauris vitae orci maximus ultricies. Cras fringilla ipsum magna, in fringilla dui commodo a.

Etiam vehicula luctus fermentum. In vel metus congue, pulvinar lectus vel, fermentum dui. Maecenas ante orci, egestas ut aliquet sit amet, sagittis a magna. Aliquam ante quam, pellentesque ut dignissim quis, laoreet eget est. Aliquam erat volutpat. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Ut ullamcorper justo sapien, in cursus libero viverra eget. Vivamus auctor imperdiet urna, at pulvinar leo posuere laoreet. Suspendisse neque nisl, fringilla at iaculis scelerisque, ornare vel dolor. Ut et pulvinar nunc. Pellentesque fringilla mollis efficitur. Nullam venenatis commodo imperdiet. Morbi velit neque, semper quis lorem quis, efficitur dignissim ipsum. Ut ac lorem sed turpis imperdiet eleifend sit amet id sapien.



Maecenas mauris lectus, lobortis et purus mattis, blandit dictum tellus.

Maecenas non lorem quis tellus placerat varius. Nulla facilisi. Aenean congue fringilla justo ut aliquam. Mauris id ex erat. Nunc vulputate neque vitae justo facilisis, non condimentum ante sagittis. Morbi viverra semper lorem nec molestie. Maecenas tincidunt est efficitur ligula euismod, sit amet ornare est vulputate.

In non mauris justo. Duis vehicula mi vel mi pretium, a viverra erat efficitur. Cras aliquam est ac eros varius, id iaculis dui auctor. Duis pretium neque ligula, et pulvinar mi placerat et. Nulla nec nunc sit amet nunc posuere vestibulum. Ut id neque eget tortor mattis tristique. Donec ante est, blandit sit amet tristique vel, lacinia pulvinar arcu. Pellentesque scelerisque fermentum erat, id posuere justo pulvinar ut. Cras id eros sed enim aliquam lobortis. Sed lobortis nisl ut eros efficitur tincidunt. Cras justo mi, porttitor quis mattis vel, ultricies ut purus. Ut facilisis et lacus eu cursus.

In eleifend velit vitae libero sollicitudin euismod.

Fusce vitae vestibulum velit. Pellentesque vulputate lectus quis pellentesque commodo. Aliquam erat volutpat. Vestibulum in egestas velit. Pellentesque fermentum nisl vitae fringilla venenatis. Etiam id mauris vitae orci maximus ultricies. Cras fringilla ipsum magna, in fringilla dui commodo a.

Etiam vehicula luctus fermentum. In vel metus congue, pulvinar lectus vel, fermentum dui. Maecenas ante orci, egestas ut aliquet sit amet, sagittis a magna. Aliquam ante quam,

pellentesque ut dignissim quis, laoreet eget est. Aliquam erat volutpat. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Ut ullamcorper justo sapien, in cursus libero viverra eget. Vivamus auctor imperdiet urna, at pulvinar leo posuere laoreet. Suspendisse neque nisl, fringilla at iaculis scelerisque, ornare vel dolor. Ut et pulvinar nunc. Pellentesque fringilla mollis efficitur. Nullam venenatis commodo imperdiet. Morbi velit neque, semper quis lorem quis, efficitur dignissim ipsum. Ut ac lorem sed turpis imperdiet eleifend sit amet id sapien.

MAK CS

Translators

Translators

→ Actual Programming code.

- Software's used to convert Source code into object code.
  - Machine Code
  - Ready for execution
  - Understood by CPU
  - Binary Form

1. Compiler
2. Interpreter
3. Java Virtual Machines

Compiler

1. Source Code is provided to compiler
   - Initially no data is provided.
2. Code is analyzed.
3. First Line of code is READ, converted into Intermediate Code
   - Error Free.
4. If a Line having an Error, Error is recorded and we move to the next line until whole code is completed.
5. At end of code, Contain Errors, they are reported — Error Report Generated.
6. If code having no errors, an Executable File is generated. Whole code is converted in one go.

Advantages

- Always generate an executable file
- Executable files cannot be altered since no source code is available.
- Mostly they are Error Free.
- Only Executable file shared, easy to distribute — No source code.
- Executable file can run on different Machines
- Conversion is Fast, since whole code in go.

Disadvantages

- If .exe file is not tested properly, can crash memory.
- Since the whole code is converted in one go, memory can CRASH.
- Finding Errors becomes complex

Interpreters

1. Source code along data is provided
2. Code is analyzed.
3. First Line of code is Read and Executed, if no Errors.
4. If a Line contain Error, Execution is halted
5. Same steps are performed until whole code is converted.

Advantages

1. Finding and Removing Errors is Easy
2. Cannot crash memory.
3. If source code is purchased, interpreter comes along with code.

Disadvantages

- It takes Longer time for execution
- To share code, source code is provided, alterations can be done.
- To execute code, always need interpreter.
- Code running on one machine cannot be interpreted on another.

JAVA VIRTUAL Machines

- Programs are partially compiled (BYTECODE) and partially interpreted
- Hybrid approach and offer portability due to byte code format.
- Platform independent.

- Partially compiled programs can be used on different platforms as they are interpreted when RUN
- Code is optimised as machine code is generated at Run Time.
- Source code does not need to recompiled, more efficient to RUN.

IDE --- Integrated Development Environment

- Softwares in which we perform coding.
- Programs are written and tested by IDEs
- Source Code Editor
- Compiler and Interpreter
- Debugging
- Auto documentation.

Pretty Print

- Displaying or printing well set of formatted source code.
- Easier to read and understand.
- Text colours are changed, underline as well.

Debugging :

- Finding Logical Errors by Running the code and Tracking the variables in variable window watch.

Break Points

- A deliberate pause in the execution of a program during testing so that we can observe values of variables.

Auto documentor

- Provide information about different function
- How to implement certain functions.

Report Window / Watch Window

- Separate window, in Runtime that shows the contents of variables updating during execution of program.

WRITE CODE

- Editor
- Pretty Print
- Expand and collapse into different blocks
- Dynamic syntax check
- Auto formatting
- Context sensitive prompts, help to complete statements

Testing Code

- Stepping - Single
- Break points
- Debugger
- Report Windows